Qassim University
College of Engineering



Introduction to Engineering Design II

GE 213

Introduction to Microsoft Excel

Part I : Basics

Overview

This handout is intended to provide an **introduction** to developing mathematical models using a spreadsheet. The instructions will guide you through the **process** of creating a model, solving the model, and displaying the results using charts and tables. After completing this handout, you should have a sense of the power and flexibility of a spreadsheet program (specifically MS Excel) and be able to create your own spreadsheets. Only the most basic spreadsheet commands/operations are used in this handout. Most of the operations illustrated can be accomplished by using alternate, or short cut, methods and you will no doubt use these other methods as you become more familiar with Excel. This handout can be **supplemented** with using the **Help** function in Excel.

Reflection is an important process in learning. Based on the experience of prior students, you should pause at the end of **each part** of any given tasks to make notes on the important points of the tasks or operations you have just completed (i.e., what do you think was important?). After completing your 'reflections,' you might also want to try to perform the operations/tasks that you have just completed **without looking** at this document. These practices are encouraged in GE 213 in order for students to become **life-long learners.**

Suggestions for Using Other Versions of Excel

In this handout MS Excel 2007 will be used and referenced. If you are using an **earlier version** of Excel at home (e.g., Excel 97-2003), you can save your files on campus as an earlier version. To do this, select **Office Button** on the menu bar, then pull down to **Save As**. A pop up window will appear. In the box beside **Save as:** you can select the version of Excel you have at home. Note that in coming from home to school (i.e., from an earlier version to the more recent version), you don't have to do anything special. The newest versions of programs will read all the older ones, but not vice versa.

Notes on the Computing Environment

In this handout an assignment will be given to you. This assignment is primarily intended to help you learn to use the Excel program which will be used in Modeling.

Excel is a **Windows** program; it is compatible with other Windows programs (e.g., Word) and is used in the Windows environment. This document does not provide Windows instructions. It is highly recommended that you familiarize yourself with Windows. This familiarity will help you use Excel as well as any other Windows compatible software you may find useful in this and other courses.

Getting Started: Mouse Tips

The ASU personal computers are all equipped with a 'mouse.' You should familiarize yourself with its use, particularly in Excel since the shape of the mouse pointer in Excel changes for different actions. The following mouse 'pointer' symbols should be recognized:

- The thick white cross ✚ will appear as you move around the cells of the document window.
- The black cross hair ┼ called the fill handle, appears when you bring the pointer to the little square on the lower-right corner of the current cell. It is generally used during copying and sizing operations.
- The thick white arrowhead ↖ is the pointer used on the toolbars, menu bars, and control boxes along the edges of the window.
- The I-beam ⌶ is used in editing operations.
- The hour glass ⧖ symbol tells you that the program is working; wait for the next prompt.

The right button on your mouse is very useful for Excel operations. Click the right hand button once and a pull-down menu of the possible Excel operations will appear. Using the right hand button for this purpose minimizes the need to use the menu bar. The use of this and other short cuts can substantially reduce the time required to develop an Excel spreadsheet model.

Getting Started: Display Screen

When you log on the PC's, the desktop screen will appear. Move the mouse pointer (the arrow) to the **Start** button in the lower left corner of the screen. Click the left mouse button, then click **All Programs**, then **Microsoft Office**, and finally **Excel**. Following Excel screen will appear:

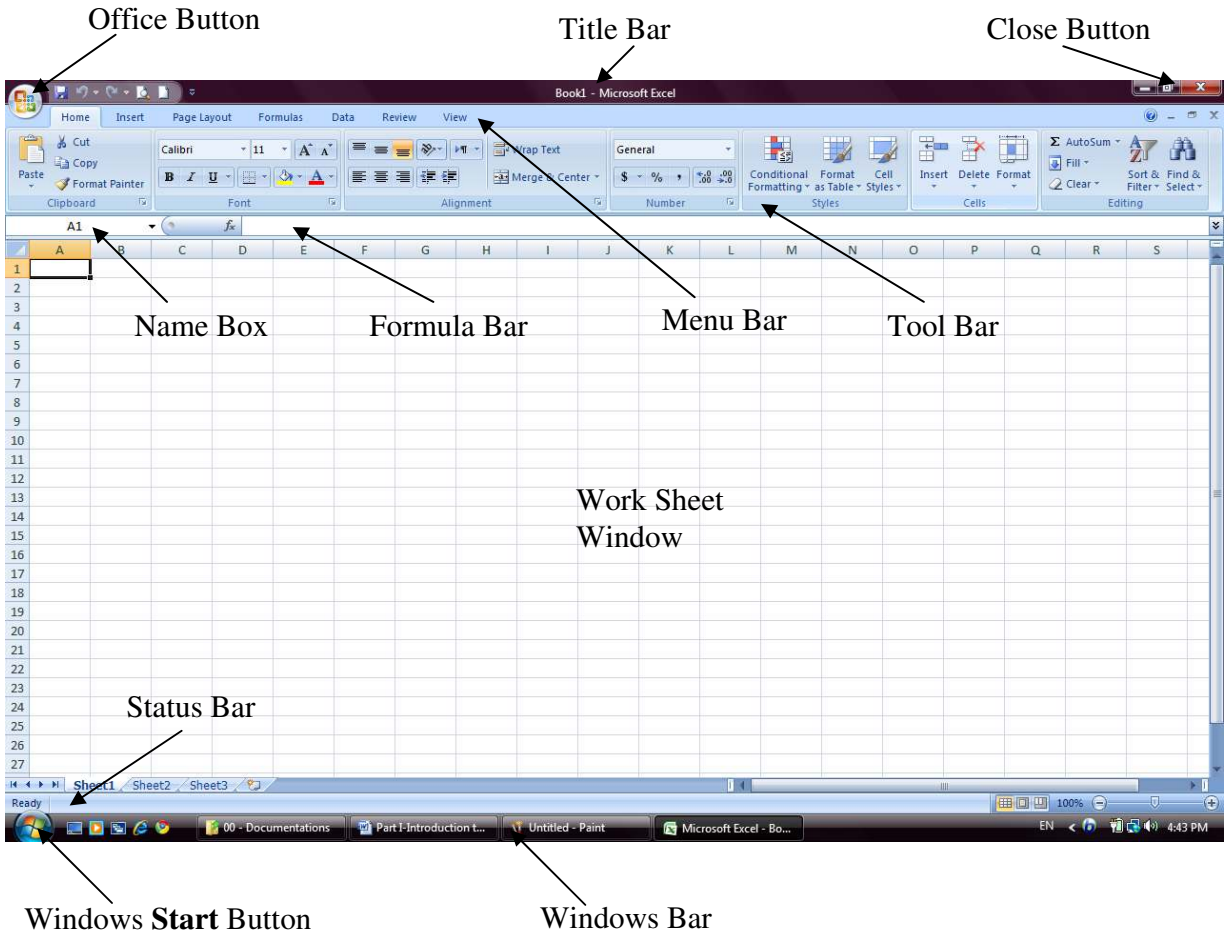


Figure 1 Excel startup screen

Note the following key parts of the Excel program window (as illustrated in Figure 1):

- **Title Bar** – the top most bar on the Excel window; it shows the name of the file that is currently running
- **Close Button** – the X in the upper right hand corner; it is used at the end of the session to close the file (bottom X) or the program (top X)
- **Menu Bar** – lists the main Excel pull-down menus; these menus allow you to select from a variety of commands used in Excel
- **Tool Bars** – short cut methods for various commands; if you point the mouse at these boxes, a highlighted box will appear which describes the function of the box
- **Name Box** – displays the address or name of the current cell

- **Formula Bar** – displays the contents of the current cell; it shows the text, data, or formula that has been entered in the current cell
- **Work Sheet Window** – this area is divided into cells where relevant information is entered; the cells are addressed by a column letter and row number (i.e., B5); the column letters are shown along the top of the document window and the row numbers are along the left hand side of the window
- **Status Bar** – displays information about the current status of Excel and may include messages relating to the activity that you have chosen to perform; the right hand portion indicates some of the settings of the keyboard (i.e., Caps Lock…)

In addition, the Excel window has some standard Windows control buttons/bars: Scroll Bars, Scroll Arrows, Minimize Button, Size Box. You should become familiar with these control features by studying Windows (you can select **Help** from the **Start** button and execute the **Windows Tour** to get an introduction).

Getting Started: Work Sheet Document Window

As mentioned above, the Excel Document Window is divided into columns and rows. The cell is a box on the document window and can contain information: text, data, or equations (formulas). The coordinated use of this information constitutes the spreadsheet application. An Excel sheet can contain 16,384 columns and 1,048,576 rows, extending well beyond the boundaries of the monitor screen. A document may contain extra **sheets** to provide a large three-dimensional domain. These other **sheets** can be accessed by clicking on the labeled tabs at the bottom of the document window.

Getting Started: Cell Selection

When you start Excel, the **A1** cell is automatically selected as the current cell (see Figure 1). This cell is highlighted, meaning it is ready to receive input from you. You can change the current cell by moving the mouse's pointer to any desired cell and clicking the mouse. The new current cell will be highlighted and the address or name of the new current cell will appear in the **Name Box**.

Getting Started: Entering Text

Three types of information can be entered into the Excel cell: **text, data**, and **formulas**. Excel will automatically interpret what you type and select the appropriate category for the information. For example, if what you type begins with a letter of the alphabet, it **automatically** becomes **Text**. Alternatively, if the information begins with an **equals sign** (=), Excel **automatically** assumes that the information is a **formula**.

Text material is entered directly by typing the information through the keyboard. **Text** is normally '**left justified'** in Excel; it is displayed in the left-hand portion of the cell. If you wish to specify that certain information within a formula is text, you should put single quotation marks around it.

If the information is data, it usually consists of numerical values and Excel will treat it as such. Excel **data** is '**right justified'** in Excel; it is displayed in the right-hand portion of the cell. This convention can help you distinguish between text and data easily.

As mentioned above, formulas start with an equals sign and are a convenient way of calculating tables of values. We will show you how to enter formulas explicitly later.

As you type the information, it will appear in the current cell and also in the **Formula Bar**. When the information is completely typed, press Enter.

EXCEL MODELING ASSIGNMENT (to be handed in)

Entering a Descriptive Title, Your Name, and Your Team Number

For every assignment, you must enter your name and team number in your spreadsheet. Select the cell **F1** as the current cell, then type the following and press **enter**:

Your Name : Your Team Number

The standards for organizing and presenting technical work require that every spreadsheet contain a descriptive title. Select cell **A1**, then type the following and press **enter**:

NASA Cone Design problem

Note the lower case 'p' in the word problem. Notice that the title you just entered overflows into cells **B1** and **C1**; this should not present a problem unless you wish to enter some other information into these cells.

Editing the Descriptive Title (Editing Text)

Text that has been entered into a cell may need to be corrected or erased. The simplest way to correct the entry, if the text is not lengthy or complicated, is to retype it. To do this, select the cell you wish to edit. The old text will appear in the **Formula Bar**. Typing will overwrite the old text.

Another way to edit text is to make correction on the **Formula Bar**. Select the cell in question. Move the mouse's pointer to the point in the text in the **Formula Bar** where you wish to edit (the cursor symbol will turn into the I-beam). We want to capitalize the 'p' in problem. Select cell **A1** as the current cell (even though the word 'problem' appears in cells **B1** and **C1**, remember that **A1** is where you entered the title). Move the pointer to the **Formula Bar** just before the letter 'p' and click the mouse (the I-beam should appear in front it). Hit the **Delete** button and the letter 'p' is erased. Now type a capital letter 'p.' While here, notice that you can move the I-beam to the left or right by pressing the appropriate **arrow keys**. This can make editing easier when you have multiple errors to fix. Press **Enter** to keep your changes to the title.

To delete text from a cell, select that cell as the current cell. Click the mouse right hand button once, menu of the possible Excel edit operations will appear. Holding the left mouse button down, move the cursor down to the entry **Clear contents**, the pull-down menu should like the figure below.

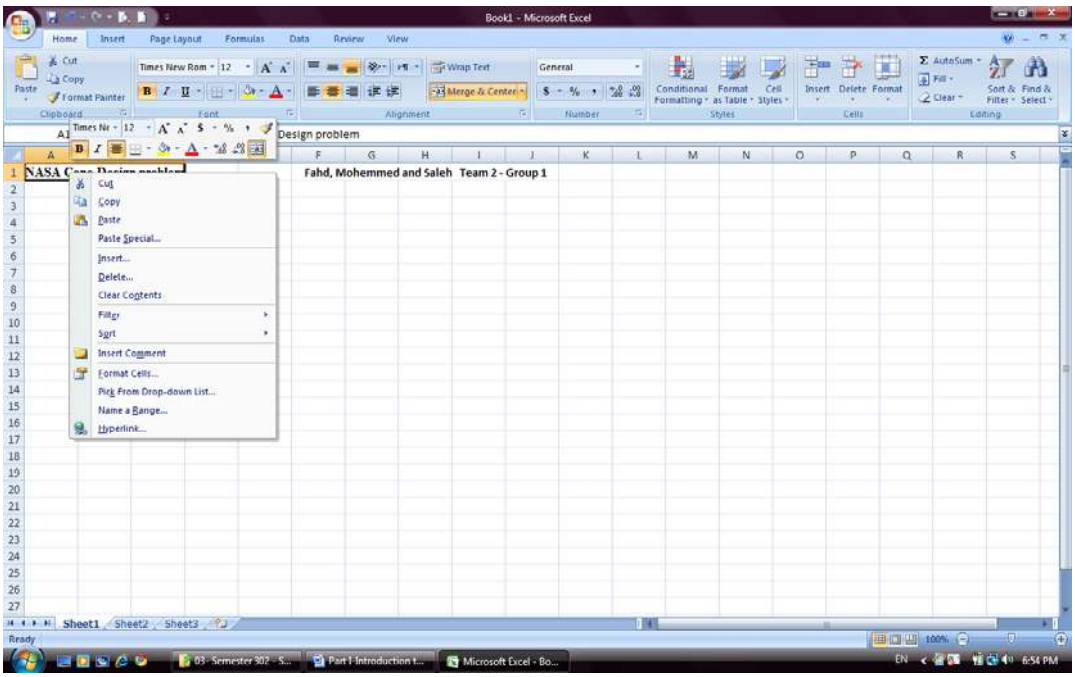


Figure 2 Edit pull-down menu

The 'Data Entry' Area, Entering & Labeling the Model Variables (including units)

The data used in a model are often referred to as 'variables,' 'design variables,' or 'parameters' and must be presented in a data entry area or section of the spreadsheet. The process for constructing this data entry area is presented below.

Re-enter the title into cell **A1** if you erased it in the previous exercise. Select cell **A4** as the current cell. Type the word **Design** and press enter (the current cell is now **A5**). Enter the word **Variables**. In cell **A7**, type **Height (cm)** and in cell **A8**, **type Radius (cm)**. Your worksheet should look like Figure 3 below.

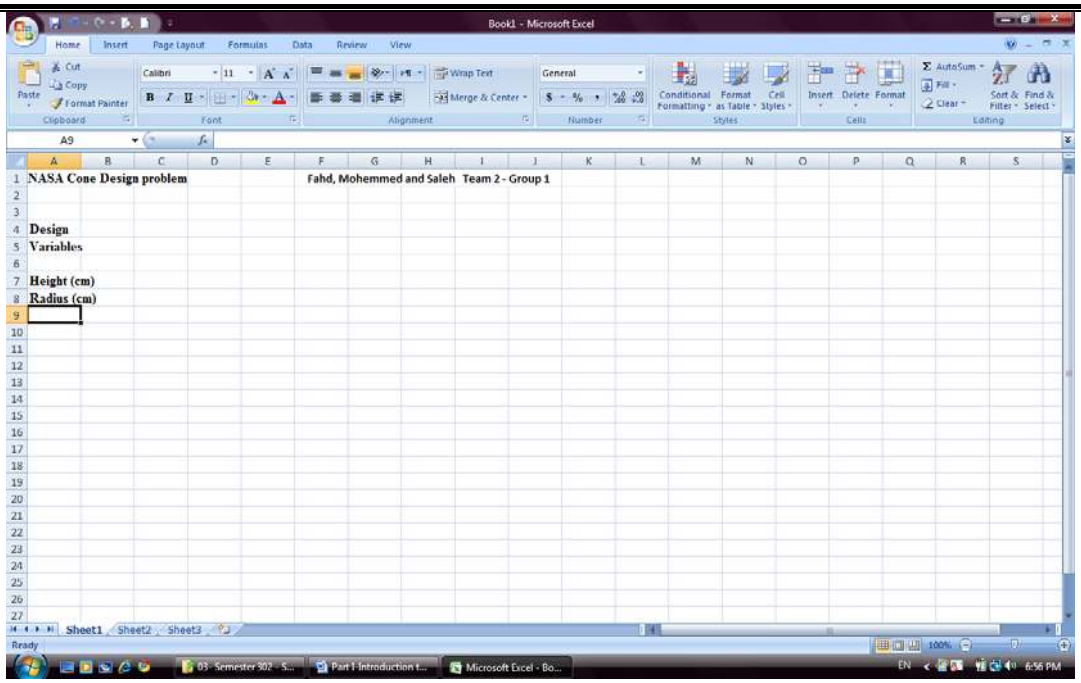


Figure 3 Excel work sheet with data entry fields

Formatting Cells

You can see that the last two entries overflow their respective cells into column B since the text is longer than the original 8.43 '**standard characters**' allocated by Excel for each cell. We are going to enter information into cells **B7** and **B8**. The overflow text would not be visible if we did not enlarge the cells so the entire text would fit within them. We are going to enlarge the entire Column A.

Since we want to adjust the cell widths to fit the contents of cells **A7** and **A8**, select cell **A7** as the current cell. Move the pointer to the **Menu Bar** and point it to the word **Format** and click the mouse. An options menu will appear. We want Excel to decide how wide the cell should be, so move the pointer to **AutoFit Column width** (see Figure 4) and click the mouse once. If you have a specific width in mind you could have selected **Column Width** and entered the size.

Your window should look like Figure 5. Note that the text in cell **A1** is still overflowing into the adjacent cells. Since we're not going to enter any information there, it doesn't matter.

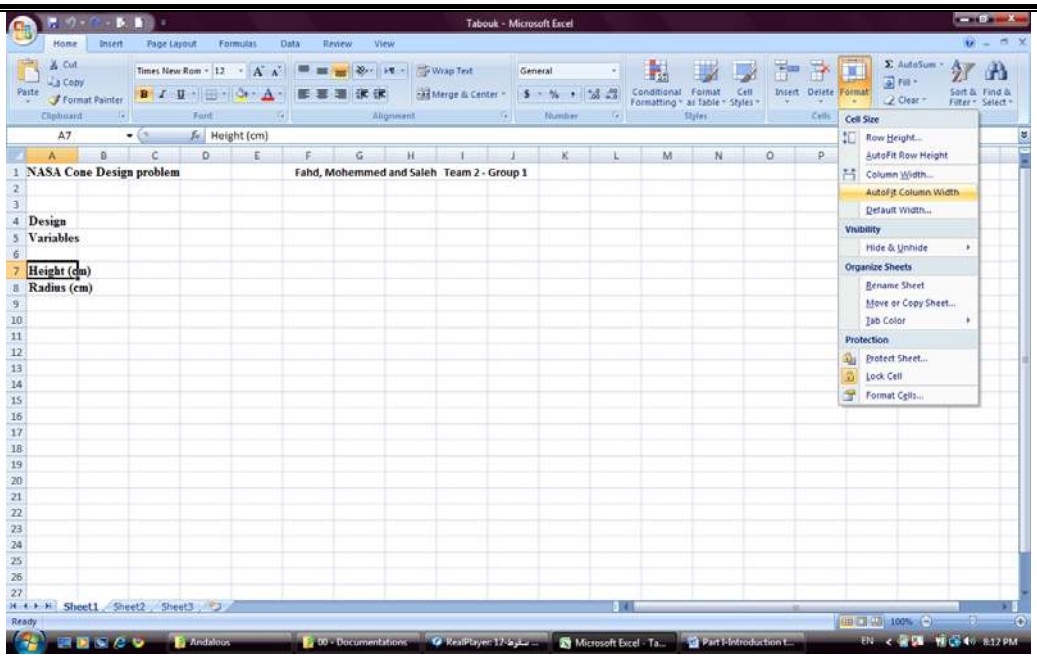


Figure 4 AutoFit selection

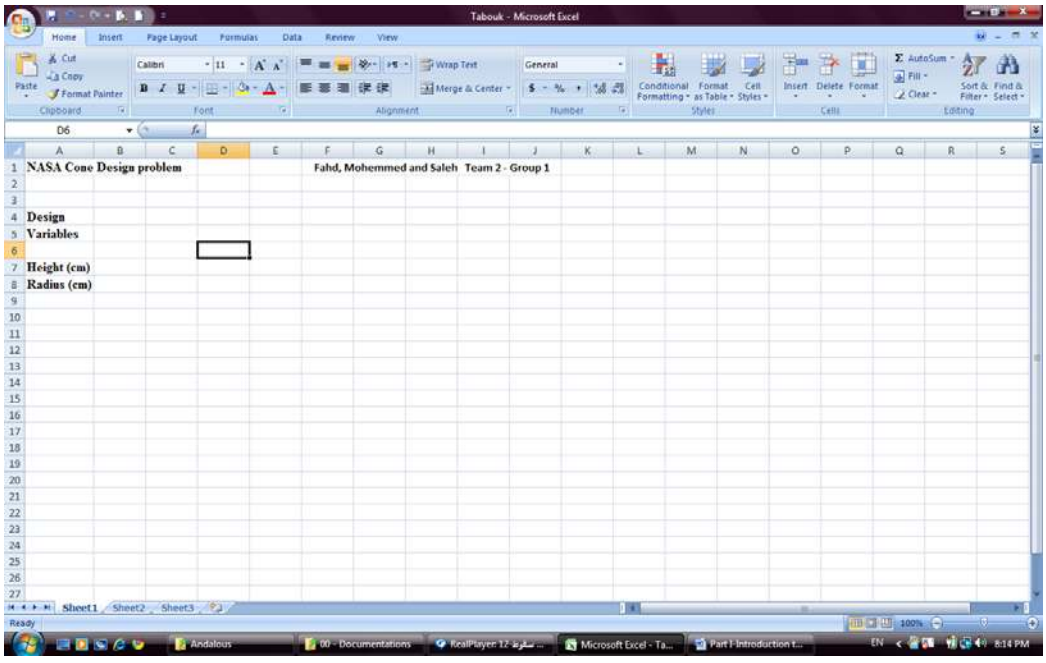


Figure 5 Work sheet after the AutoFit selection

The 'Calculation' Area, Constructing & Labeling the Model

The calculations that are required for the model must be organized and presented in a section or area that is separate and distinct from the data entry area. Select cell **F3** and enter the word **Cone**. In cell **F4**, enter the word **Radius** and in cell **G4**, enter the word **Volume**. To indicate the appropriate units, in cell **F5** enter **(cm)** and in cell **G5** enter **(cc)**.

Copying Text

There is a shortcut method for copying text into adjacent cells. We want the word **Cone** to appear in cell **G3**. To do this, select cell **F3**. Then, bring the pointer to the bottom right corner of the cell (there is a little black square in the highlighting there). The pointer will change to a **black cross** when you are in the right spot. Hold down the mouse button (make sure the **fill handle** remains) and drag the mouse across to cell **G3**. As you move the pointer into the new cell area, a fuzzy grayish highlight appears around the cell and the cell area is highlighted. Let go of the mouse button and the contents of cell **F3** are copied into **G3** (see Figure 6). Note that you can copy the text into more than one cell by continuing to drag the **fill handle** across. You can only copy in one direction. If you want to copy into a second direction (down, for example), you must repeat the process.

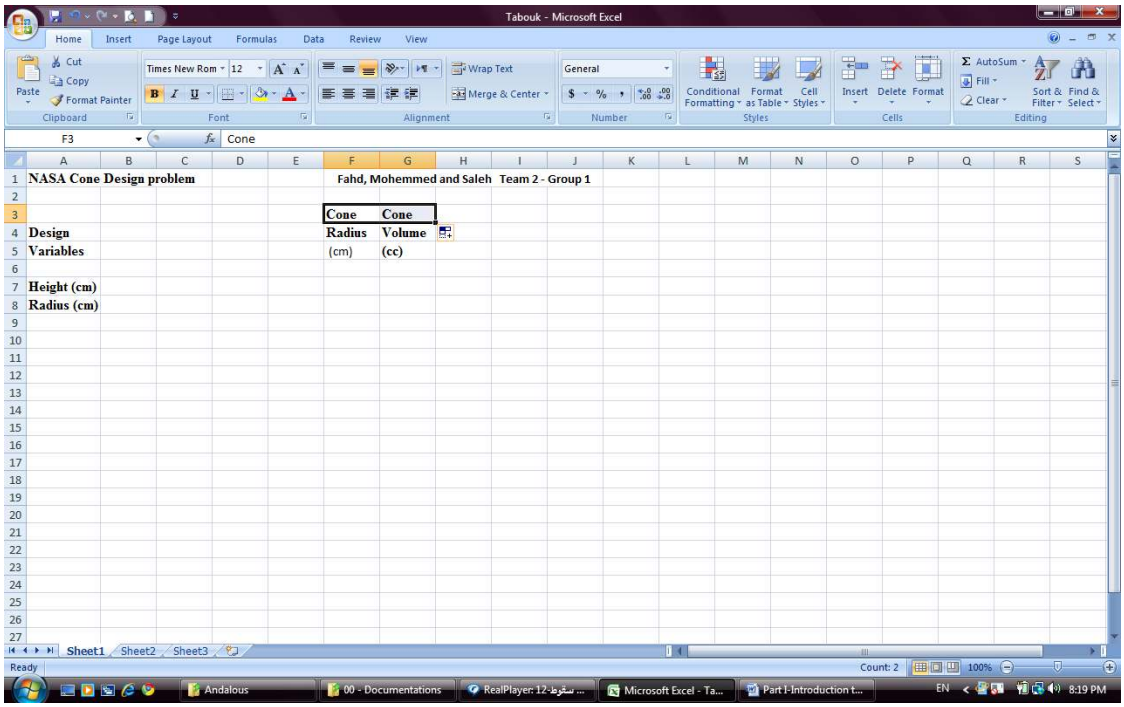


Figure 6 Copying text

Entering Data

You must organize your spreadsheet in such a manner that it be easily read and understood by others. On the left-hand side of your spreadsheet, you prepared a data entry section. Here you can enter the input data (or constants) you will need in your calculations and can readily see the initial conditions. Furthermore, should you want to use these input data in more than one equation, you can refer to them using their cell addresses rather than rewriting the data every time. Also, should you wish to change one of these values, you only have to change it once in the data entry section instead of in every equation or formula in which it was used.

Select cell **B7** and enter the number **20** (notice that the data is right justified). In cell **B8**, enter the number **5**.

Entering Formulas (in the 'Calculation' area of the Model)

You prepared the headings for a table which will include the results on the right hand side of the spreadsheet (columns F and G). This area is called the 'calculation' area of your model. You will eventually want to evaluate cone volumes for several values of the radius; thus, you will want to tabulate your results. As the first entry in the table, select the value of the radius you entered in the data entry section (cell **B8**). You can enter the radius value into cell **F7** by typing 5, or you can equate the table entry **F7** with the original input data entry **B8** by entering the formula:

=B8

which is equivalent to saying that the content of cell **F7** equals the content of cell **B8**.

To accomplish this, select cell **F7** as the current cell and you can type =B8 and press **Enter**, or you can type = and then click on cell **B8** and =B8 will appear in the **Formula Bar** and then press **Enter**.

When you enter this formula, the numerical value 5 will appear in cell **F7**. Experiment a bit, enter a different value into cell **B8** and see how the value in **F7** changes automatically. Note that when cell **F7** is the current cell, the **Formula Bar** displays the formula while the cell **F7** displays the value 5 (or whatever is in **B8**). Now make sure cell **B8** has a value of 5 in it.

Now enter the formula to evaluate the cone volume into cell **G7**. Select **G7** as the current cell and type

=(1/3)*pi()*B7*F7^2 and press Enter

The function **pi()** represents the commonly used geometric values of 3.14... (More about functions later). In your formula you need the cone height, a value you previously entered into cell **B7**, and the square of the radius, **F7^2** (the **carrot (^)** notation represents raising to a power;

i.e., in our formula we're raising the contents of **F7** to the second power). The value 523.5988 should appear in cell **G7** (see Figure 7).

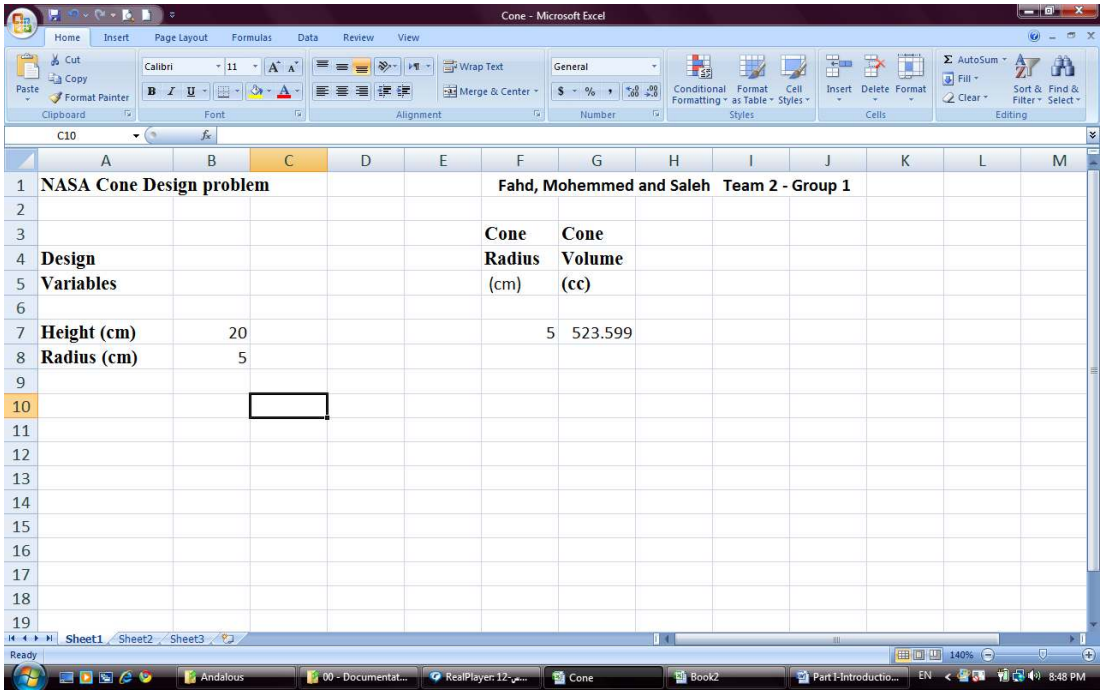


Figure 7 Entering formula

Saving Programs

You will want to save your model for future editing or use. If you are working on a lengthy problem or project, it is sound practice to periodically save your work. To save your program, move the pointer to **Office Button** and click with the left mouse button, the options menu will appear, click the **Save** or **Save as** and let up on the mouse button. The following dialogue box will appear (see Figure 8). Note: you can also get to the same pop up window by clicking the box that looks like a diskette on the **Toolbar**.

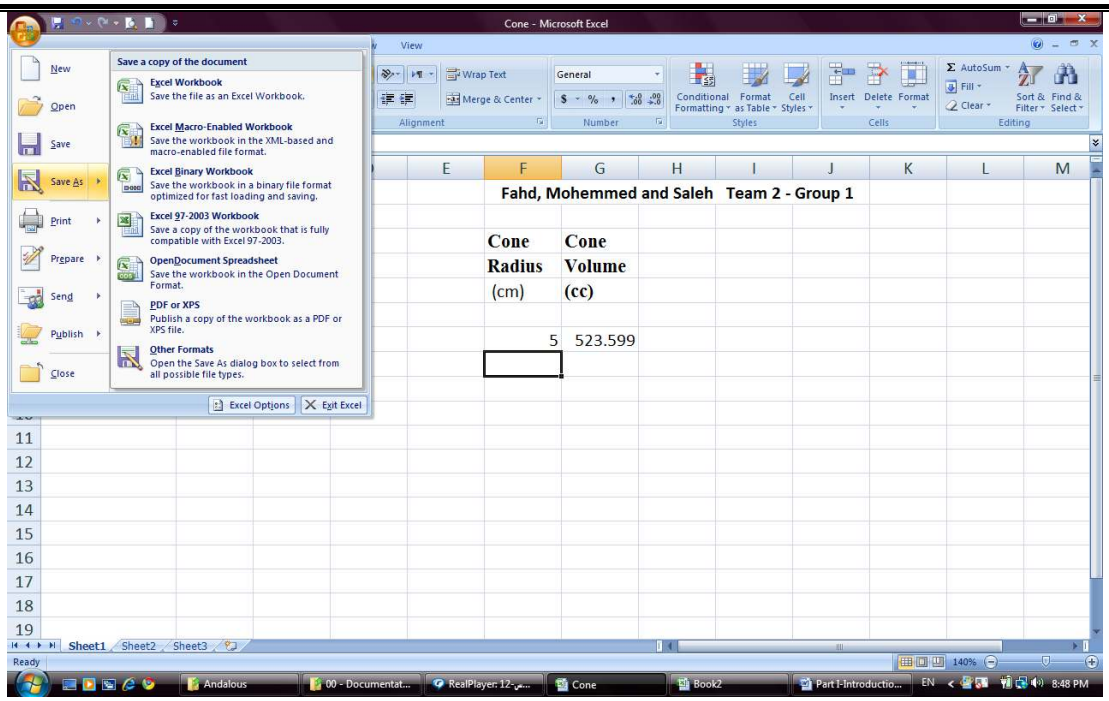


Figure 8 Saving the program

Now you may quit your session by exiting Excel by clicking on the **Close Button** on each screen. To terminate your Windows session, from the **Start** button, choose **Shut Down**. Be sure to remove your disk from the disk drive and label it.

Restart the Program

Should you wish to continue your work on a previously saved program, enter Excel via the **Start** button menu then click on **Office Button** and click with the left mouse button, the options menu will appear, click the **Open** and let up on the mouse button. The following dialogue box will appear (see Figure 9), then pick your file.

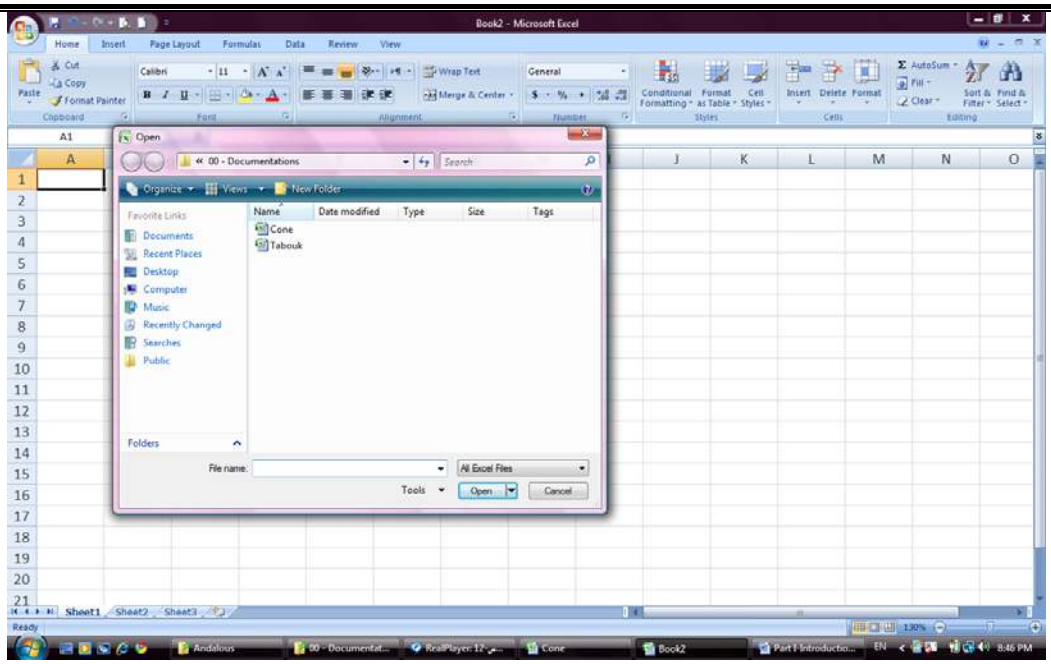


Figure 9 Open the program

Building a Table

You are now ready to build a table which shows how the cone volume varies with base radius at a fixed value of the cone height. The cone height is referred to as a model **parameter** while the base radius of the cone is referred to as the **design variable** in this exercise. A table should be as flexible as possible, so that it will be easy to change the values of the design variable and to plot the data.

You already have an initial value of the radius (entered in cell **B8**) which is automatically transferred into cell **F7**, the first cell of our table. You could manually enter other values of the radius into the table, but there is an easier method since we are going to increment the radius by a constant value. First we need to define the increment value in the input data section of our spreadsheet. Select cell **A9** and enter **Radius Inc**. Then select cell **B9** and enter the number **2**. We are now going to use a formula to increment the cone radius. Select cell **F8** and enter the formula: **= F7+B9**. If you select cell **F8** again, the formula can be seen on the **Formula Bar** and the value (i.e., **7**) can be seen in the cell (see Figure 10).

This formula could now be repeated down the line of the table. It would be rather tedious to re-type the equation eight more times, so we're going to use Excel's formula copying feature.

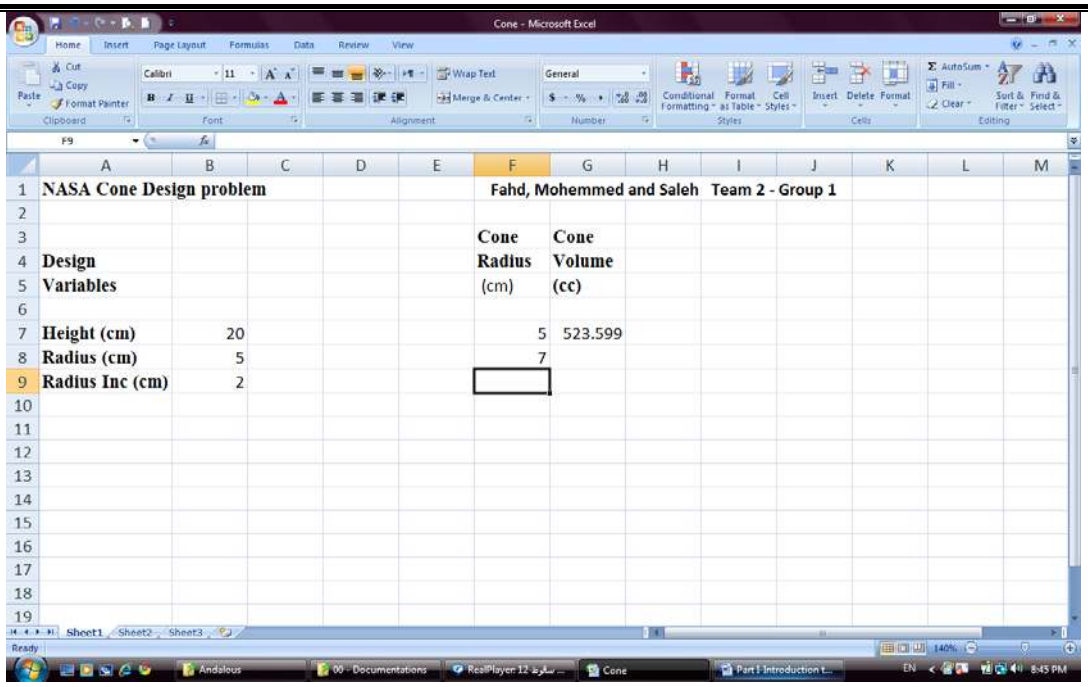


Figure 10 Building the table

Entering Relative and Absolute Addresses

It is now necessary to introduce the concept of relative and absolute addresses in a spreadsheet. **Absolute addresses** are used for data in general and for design variables or parameters in particular. We don't want to copy the formula in cell **F8** exactly or we'd get the value **7** over and over again. What we want is to copy a formula that says, "This cell equals the cell above plus the increment." So, we want the first cell address in the equation (**F7**) to change, but the second cell address to remain the same (**B9**). Excel's formula copying feature has an updating capability which will enter the appropriate formula into the cells below. This is the **relative address** feature of Excel.

There is a slight complication. Excel updates all relative addresses and not just the one we are interested. This means that if we copied the formula to cell **F9**, it would read: **=F8+B10**. To make the increment address fixed, we need to assign it an absolute address in the formula.

Absolute addresses are defined by the use of a dollar sign preceding the address. To specify the absolute column, you would put **$** in front of the letter B. To specify the absolute row, you would put **$** in front of the number 9. To make the cell absolute, you would type **$B$9**. To update your spreadsheet, select cell **F8** and edit the formula to read: **=F7+$B$9**.

Alternatively, you can give cell **B9** a name and the name will always be an absolute address. To name a cell, select cell **B9**. Then click in the **Name Box** on the **Formula Bar** and type **Increment**, then press **Enter**. You could now edit your formula to read**: =F7+Increment**.

In the copying process, the absolute addresses are copied exactly as you entered them and all other cell addresses are updated to relative addresses.

Copying Formulas

Copying formulas is similar to copying text. To copy the formula in cell **F8** into cells **F9** through **F16**, select cell **F8** as the current cell and then place the pointer at the lower right corner of the cell where it turns into the **fill handle**. Press down on the mouse and drag the fill handle down to cell **F16**. Just like when you copied text, the cell areas are highlighted in black and the region of cells are surrounded with the fuzzy gray border. When you release the mouse button, the correctly calculated radius values will appear in the cells. You should now have a column of numbers **5, 7, 9…23**. You may wish to experiment and see what happens to the values in the table as you change the initial cell radius, **B8**, or the increment, **B9**. If you do, make sure you change back to the original values before moving on.

Completing the Table

You next want to copy the volume formula into cells **G8** through **G16** to have the cone volumes associated with the various radius values. Before you copy the formula in cell **G7**, a small correction must be made to make the cone height an absolute address. There is a short-cut method for doing this instead of typing in the dollar signs.

Select cell **G7** as the current cell. Bring the pointer to the formula bar and click the mouse. Using the directional arrows, move the **I-beam** so that it is between the **B** and **7** in **B7**. Press the **F4 function key** at the top of your keyboard. This will automatically change the relative address **B7** into the absolute address **$B$7**. (Repeated pressing of the **F4 function key** deletes or enters dollar signs to give all the possible combinations of absolute and relative addresses). After editing the formula, press **Enter**. Your spreadsheet should now look like Figure 11 below.

Now with the **G7** as the selected cell, copy the formula into cells **G8** through **G16** just like you did above. When you release the mouse button, your table should look like Figure 12 below.

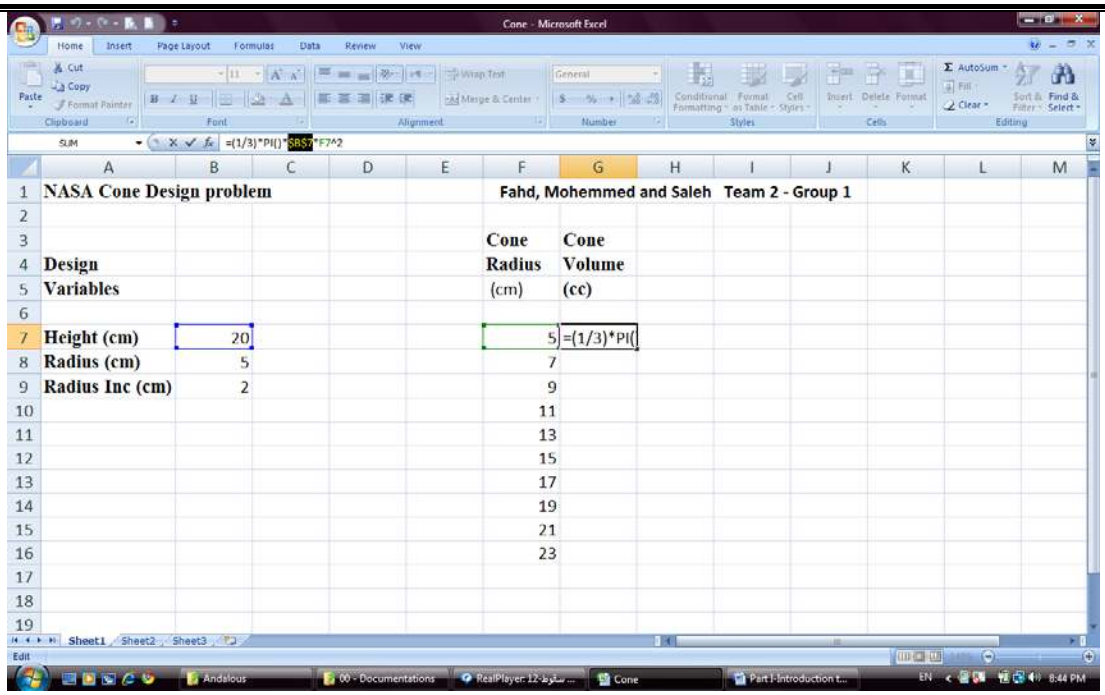


Figure 11 Using absolute addresses

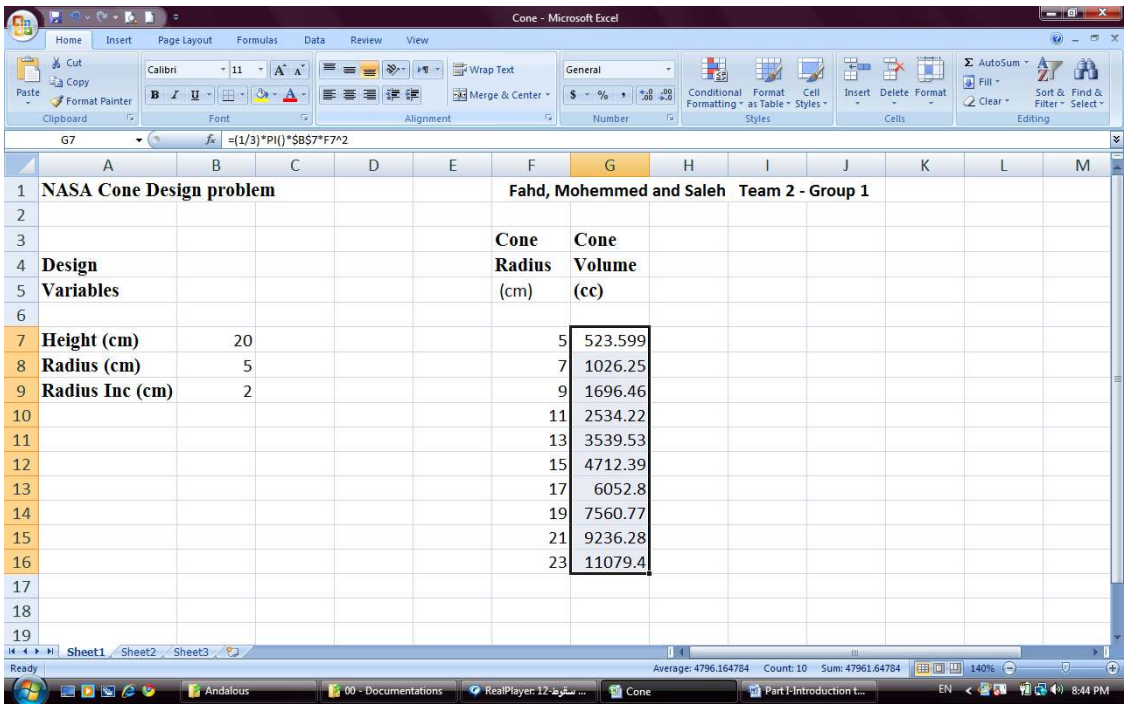


Figure 12 Completing the table

You will now want to consider the number of significant digits that you want displayed on your spreadsheet. For example, when decimal digits are not appropriate for your application, you should reduce the number to an integer. Mark the cells you wish to change (**G7** to **G16**) and click on the **decrease decimal button** on the **Home Menu Bar** and continue pressing this button until the numbers are integers (see Figure 13). Note: the actual value of the number does not change, only the number that is displayed.

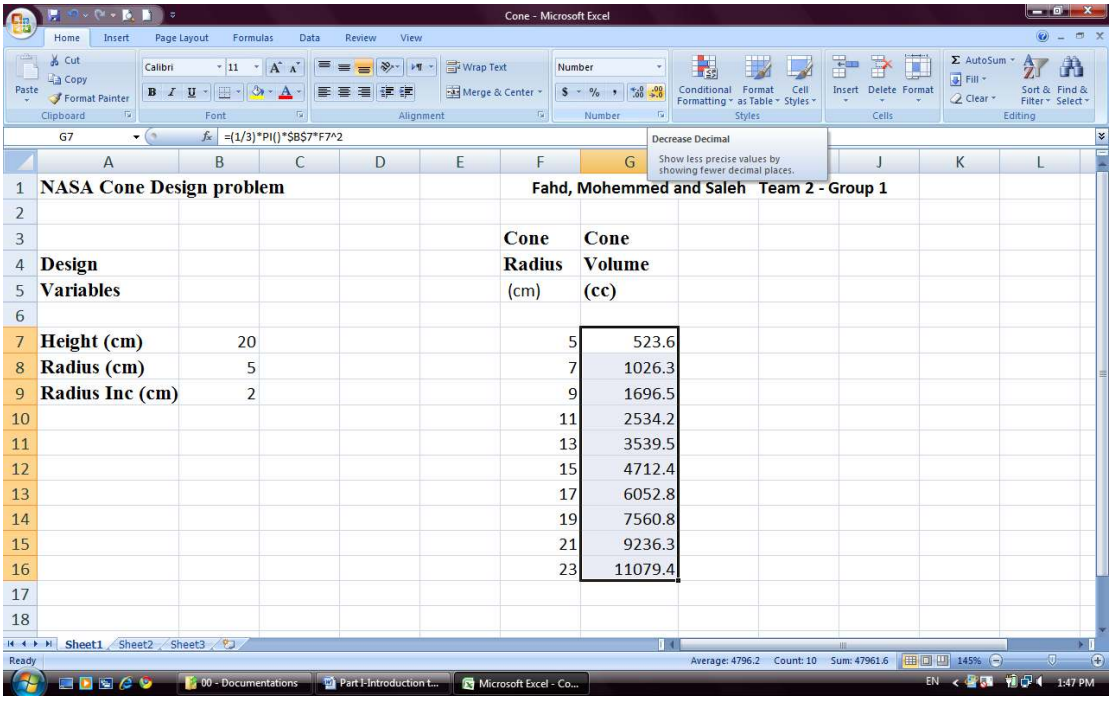


Figure 13 Adjusting decimal places

You can also change the format of your cells. If you go to **Format** on **the Home Menu Bar** and then select **Format Cells**, a pop up window appears in which you can change the Alignment, add Borders…

Creating a Chart

The next task is the creation of a chart using the data in the table you have just created. The chart should display the cone volume versus the cone radius. Excel has the capability of creating Bar charts, Line charts, Pie charts, Area charts, XY(Scatter) charts, and others. In addition, Excel provides numerous options and variations of these visual presentations. In this exercise, you will create a **XY (Scatter)** chart.

You will use **Chart Wizard** to create your chart. It is depicted by a Bar chart on the **Insert Menu Bar**. Chart generation follows a sequence of Excel controlled steps. However, at each step, there are choices for the user. You will create an embedded chart, a chart located on the same worksheet as your spreadsheet, by following the steps detailed below.

First: Select the data which you want to draw a chart for it (see Figure 14).

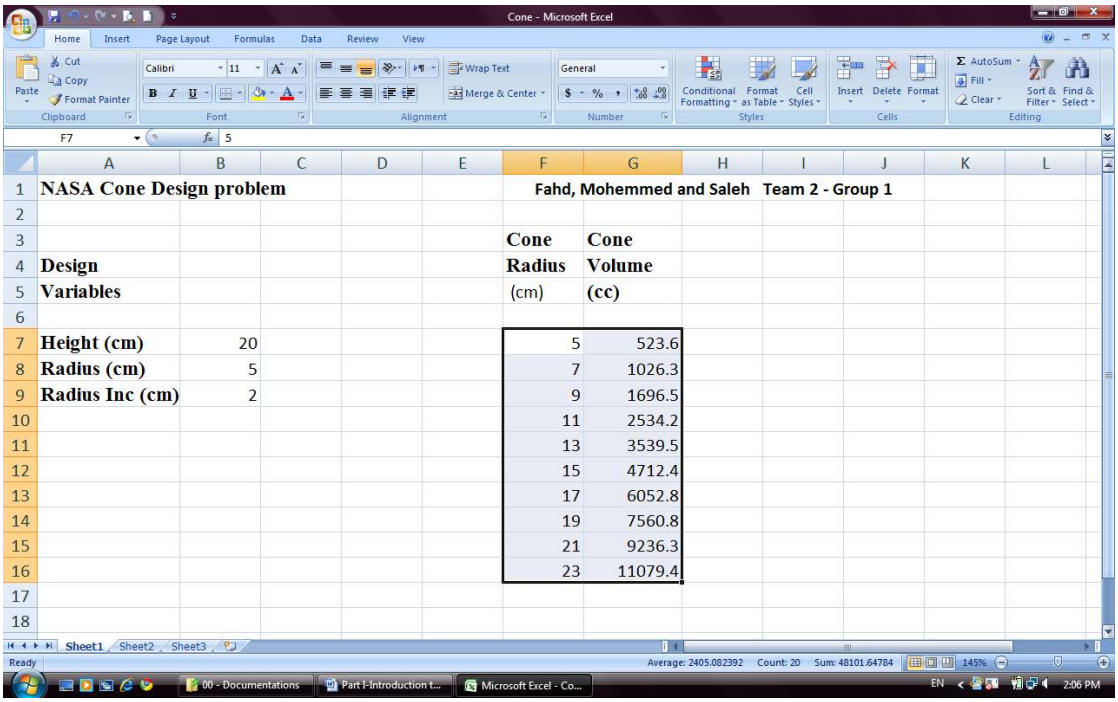


Figure 14 Selecting Data for Chart

Second: Move the pointer to the **Insert Button** and click the mouse to depict the **Chart Wizard** buttons then press on to select the type of chart you want. Select the **XY (Scatter)** Chart by clicking on it. A pop down window has Five formats of a scatter chart will appear (Figure 15A). Select the smooth curve fit format (the second format shown). A crude chart will generated as shown in Figure 15B.

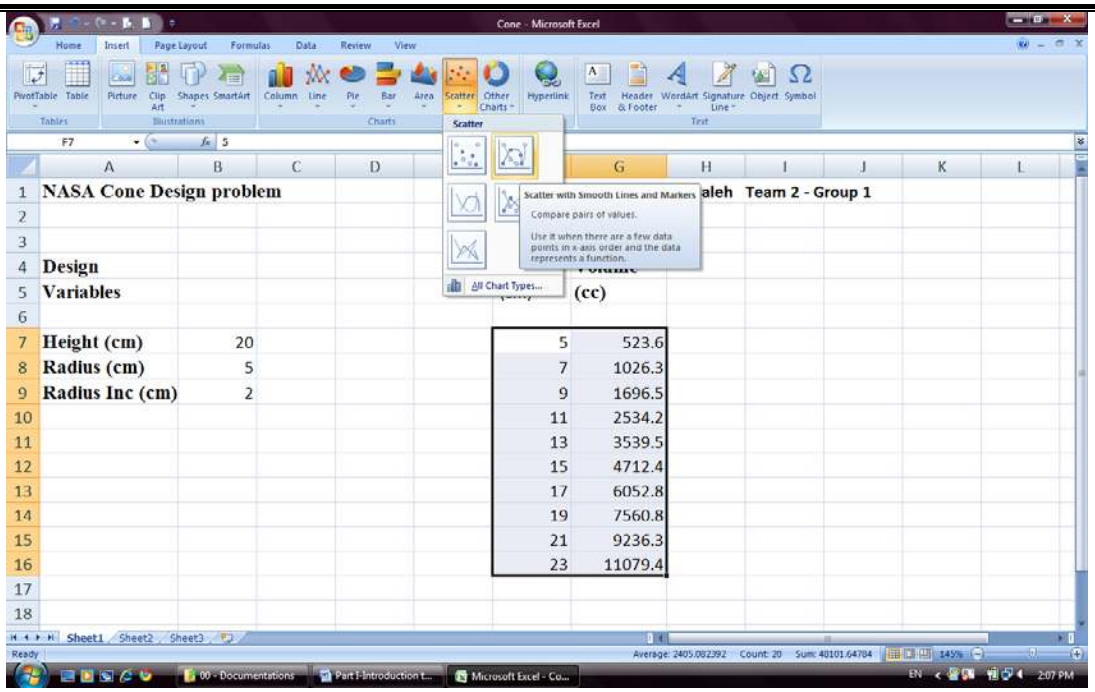


Figure 15A Chart wizard button

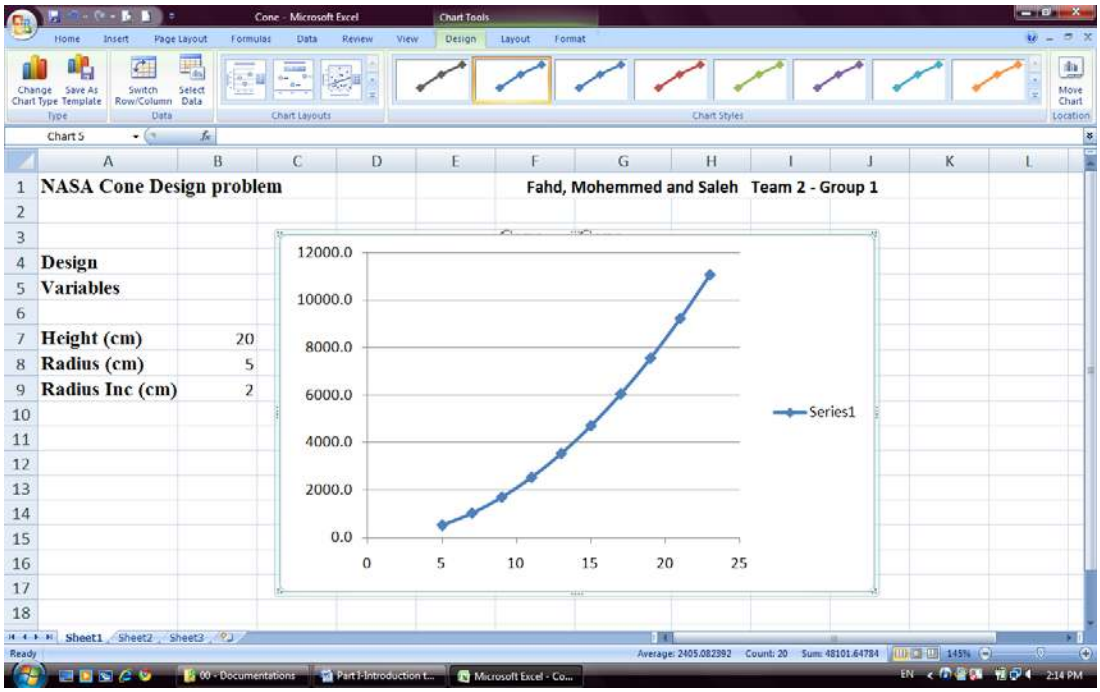


Figure 15B Crude Chart

Third: Formatting the crude: The MS Excel allows you to do formatting to the crude charts via different ways. For example we will do formatting through the following steps, however you may follow another arrangement of steps to reach the same result.

1) Formatting the size, location and the location of the chart title. This is done using the mouse as shown in Figure 16.

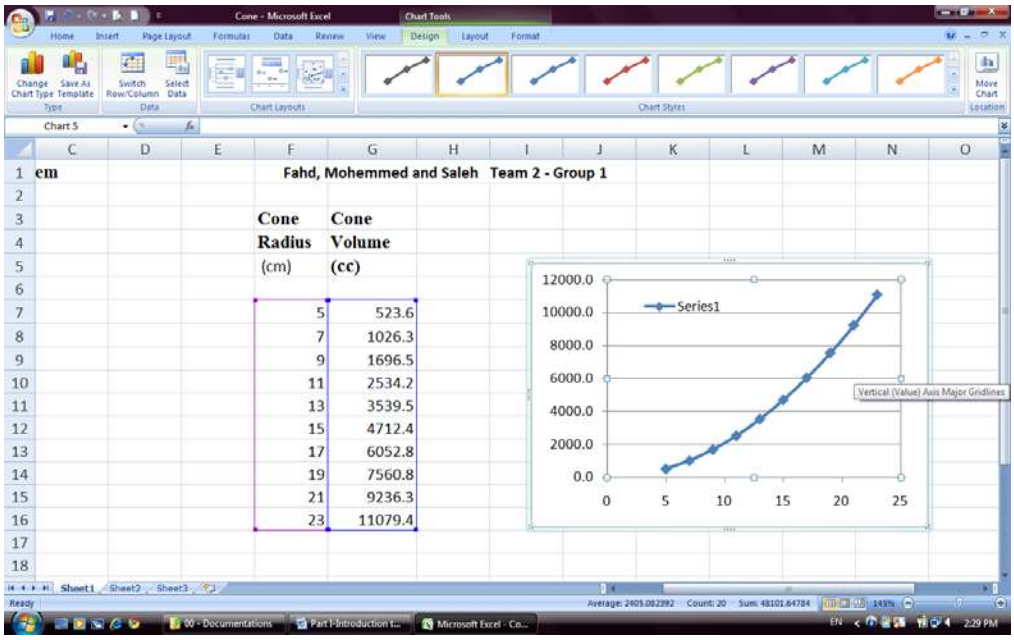


Figure 16 Formatting the size, location and the location of the chart title.

2) Formatting the plotted series: This done by choosing the **Select Data** Button from the **Design** Menu Tool Bar (Figure 17A) or by mouse right click on the plot area (Figure 17B). Both will result in appearing the **Select Data Source** menu as shown in Figure 18. This final menu allows you to do different editing option for the series like its title X data and Y data. You have to practice by yourself and try all available options then notice what changes will occur. Using these available options you can reach to Figure 19.

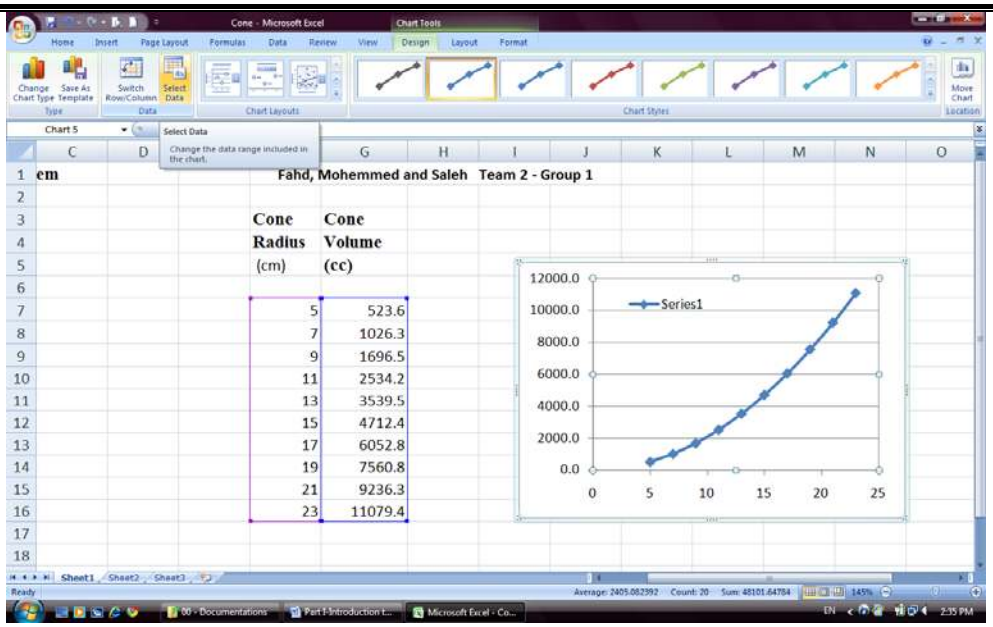


Figure 17A Choosing the Select Data Button from the Design Menu Tool Bar.

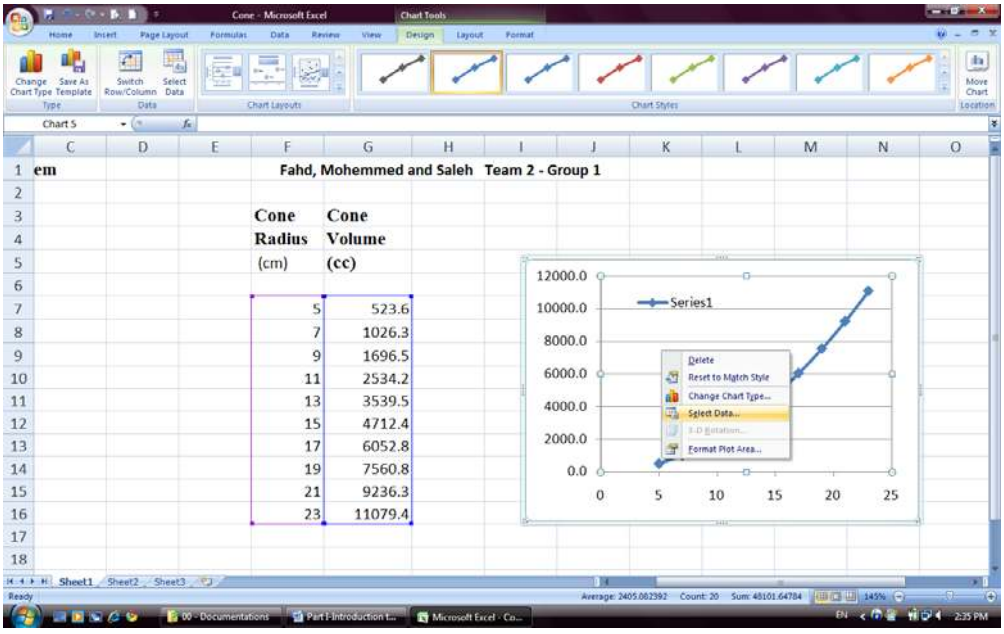


Figure 17B Choosing the Select Data Button by mouse right click.

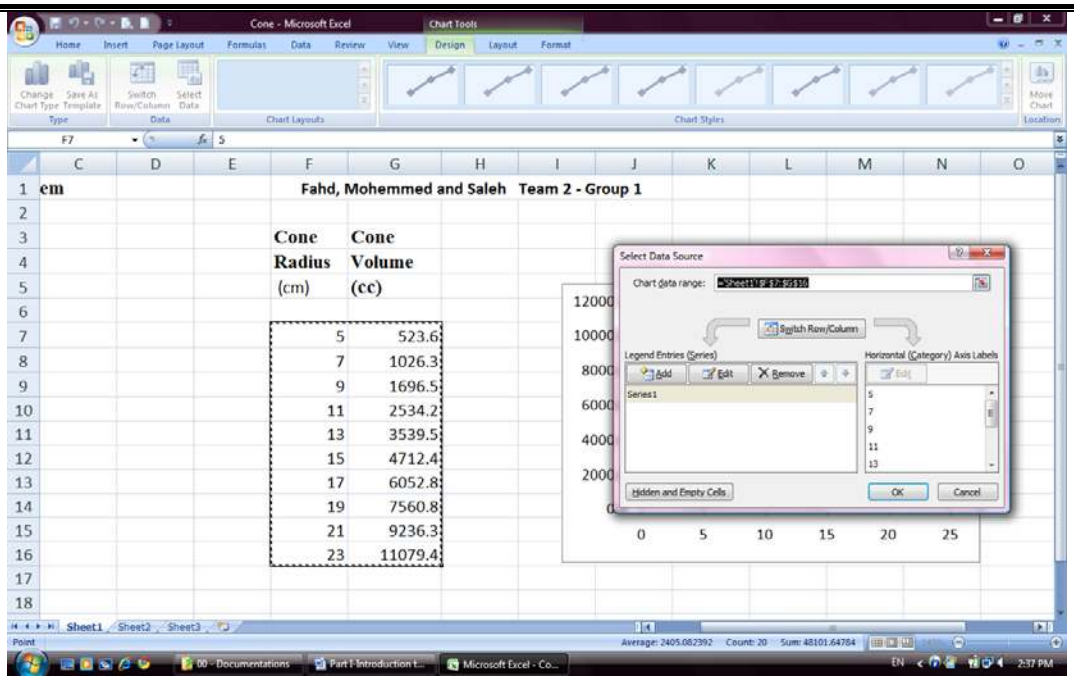


Figure 18 Select Data Source menu.

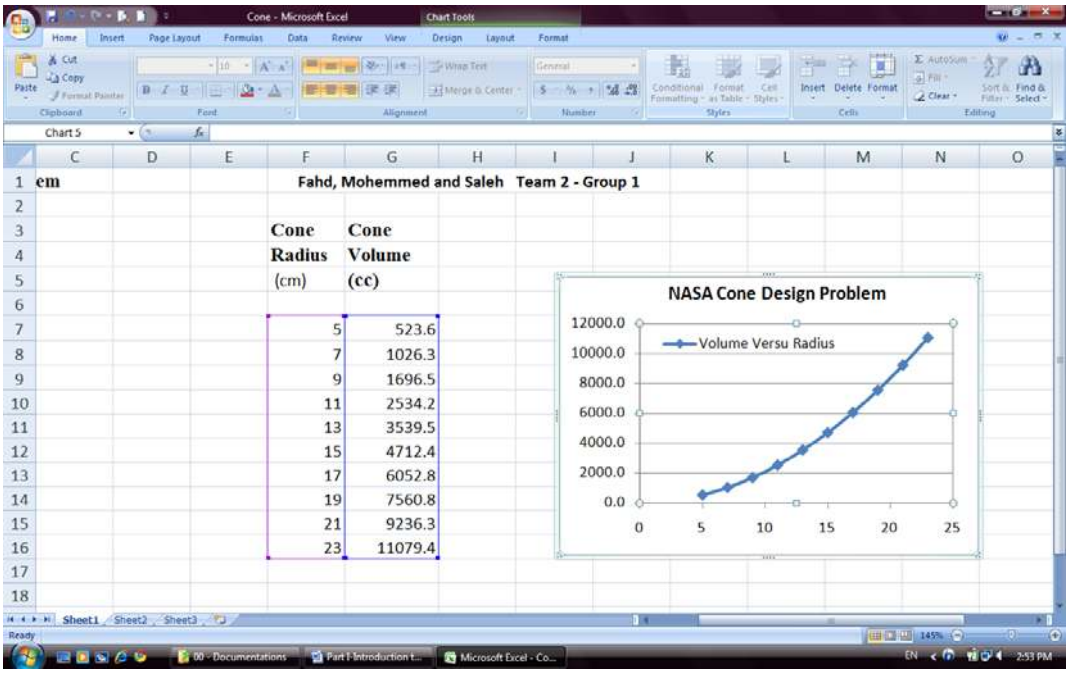


Figure 19 Plot series between volume and radius.

3) **Formatting the series curve**: You can format the series curve as follows:
   a. Select the curve using the left mouse click
   b. Right mouse click on the selected curve will result in a pop down menu
   c. Chose **Format Data Series** from this menu.
   d. You will get the formatting menu shown in Figure 20.

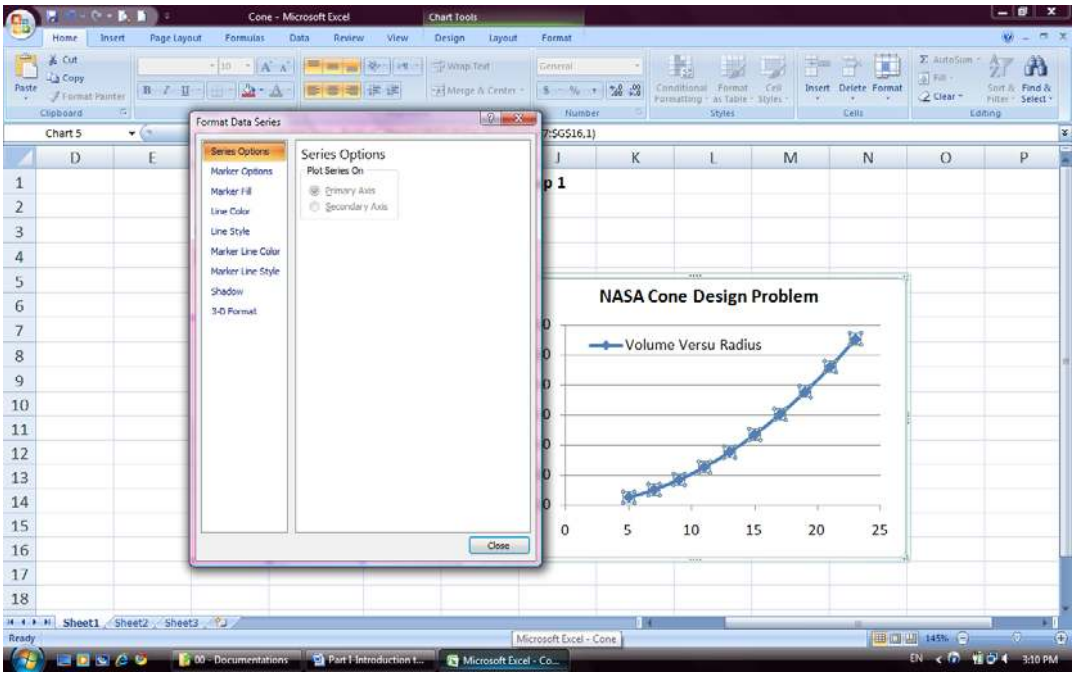


Figure 20 Format Data Series menu.

4) **Formatting the plot area:** You can format the plot area as follows:
   a. Select the plot are by left mouse click
   b. Right mouse click on the selected plot area will result in a pop down menu
   c. Chose **Format Plot Area** from this menu.
   d. You will get the formatting menu shown in Figure 21.

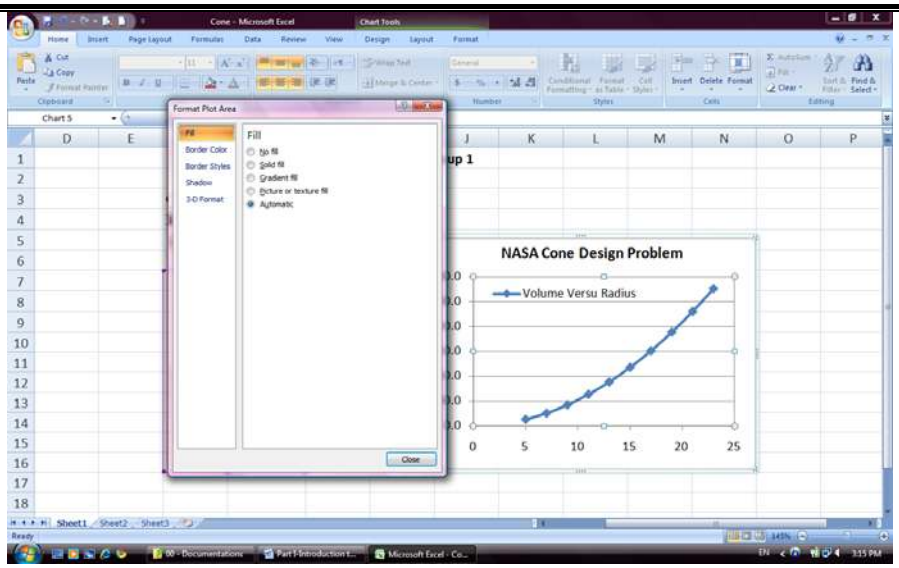


Figure 21 Format Plot Area menu.

5) **Formatting the Axis :** You can format an axis of the chart as follows:
   a. Select the axis by left mouse click
   b. Right mouse click on the axis will result in a pop down menu
   c. Chose **Format axis** from this menu.
   d. You will get the formatting menu shown in Figure 22.
   e.

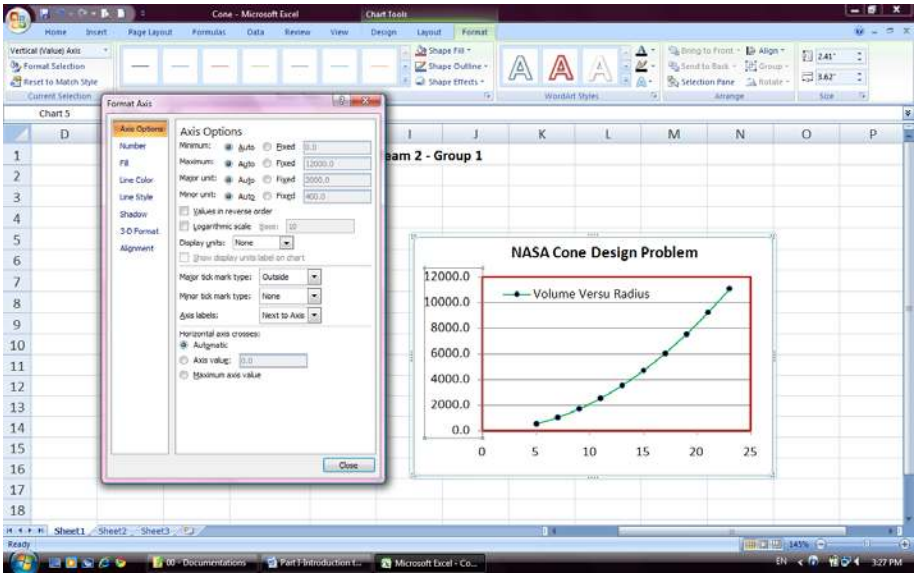


Figure 22 Format Plot Axis.

Fourth: Adding new data series: if we want to add a new data series foe the same chart, for example the surface area of the cone. Using the same procedure add a new data column for the cone surface area as shown in Figure 23. (Note this formula is used in cell H7: " **=PI()*F7*($B$7^2+F7^2)^0.5** ").

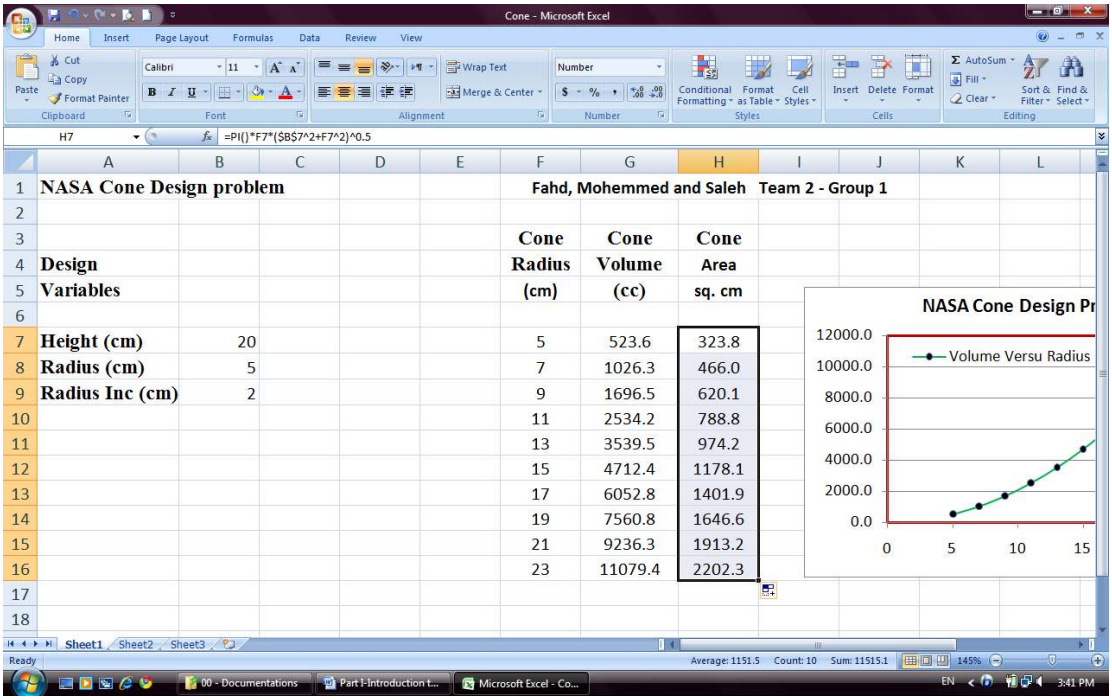


Figure 23 New data column for the cone surface area.

Then to add the new series to the chart:

a. Select the plot are by left mouse click
b. Right mouse click on the selected plot area will result in a pop down menu
c. Chose **Data Series** from this menu.
d. You will get the formatting menu shown in Figure 18
e. From this menu chose the **Edit** button a small menu will appear in which you will name the new series and define the series X and Y values (Figure 24).
f. You will get a new crude series line as shown in Figure 25.

g. This new data series can be formatted like the previous one, moreover a secondary axis may be chosen for this new data series.

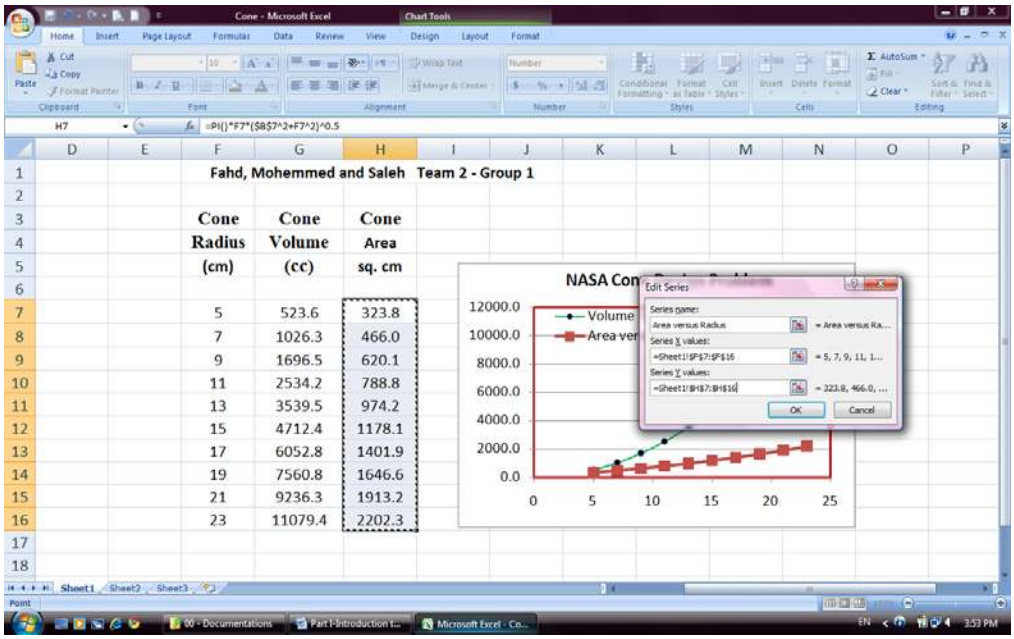


Figure 24 Add the new series to the chart.

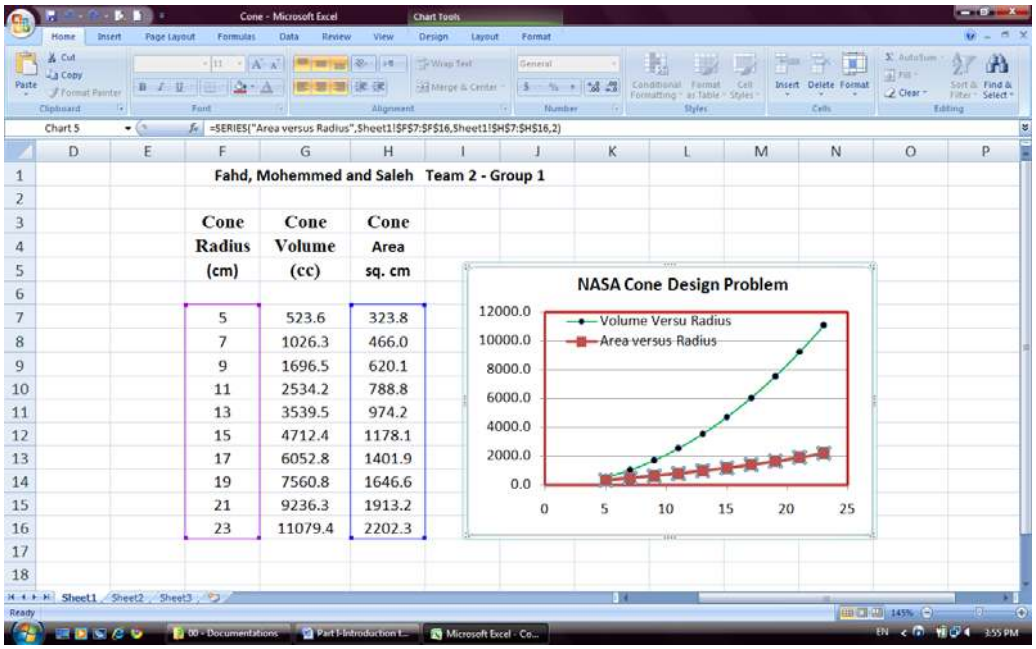


Figure 24 A crude new series.

Fifth: Finalizing a chart: using the available options in MS Excel you may finalize the chart. For example:

a. Select the curve using the left mouse click
b. Right mouse click on the selected curve will result in a pop down menu
c. Chose **Format Data Series** from this menu.
d. You will get the formatting menu shown in Figure 20.
e. Chose **Secondary Axis** to establish other Y-axis for this series.
f. Moreover do the appropriate formatting for the markers and the line.
g. Also from the Layout Tool Bar Menu you may do different types of formatting and their titles.
h. Try to reach the final form shown in Figure 25.

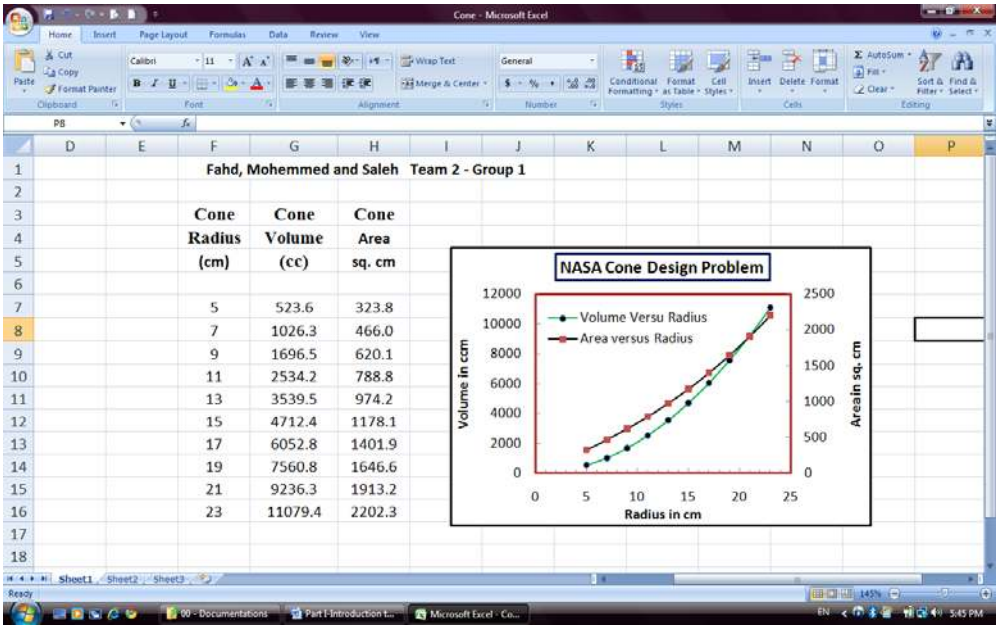


Figure 25 Final formatted chart.

Functions

Excel has numerous built-in functions to assist you with the calculations. Most of these functions will be used in your equations; therefore, they will be needed in the current cell as you enter a formula. You can select the **Insert Function** button on the **Formula Tool Bar Menu**. A dialog box will appear with two menu areas (see Figure 26A). The upper part lists the Function Category, which allows you to locate the function you need more easily. For example, you may choose the Math & Trig selection if you are looking for the cosine function (see Figure 26B). The lower part names of the specific functions. You make your selection by clicking the mouse pointer on the desired function.

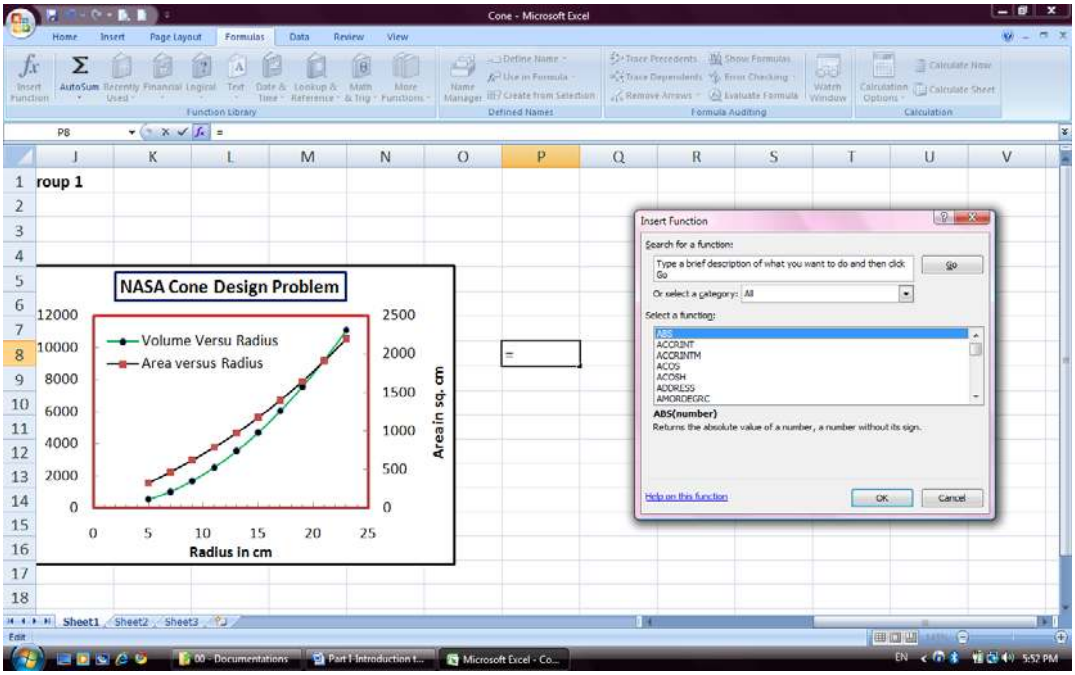


Figure 26A Function wizard dialogue box

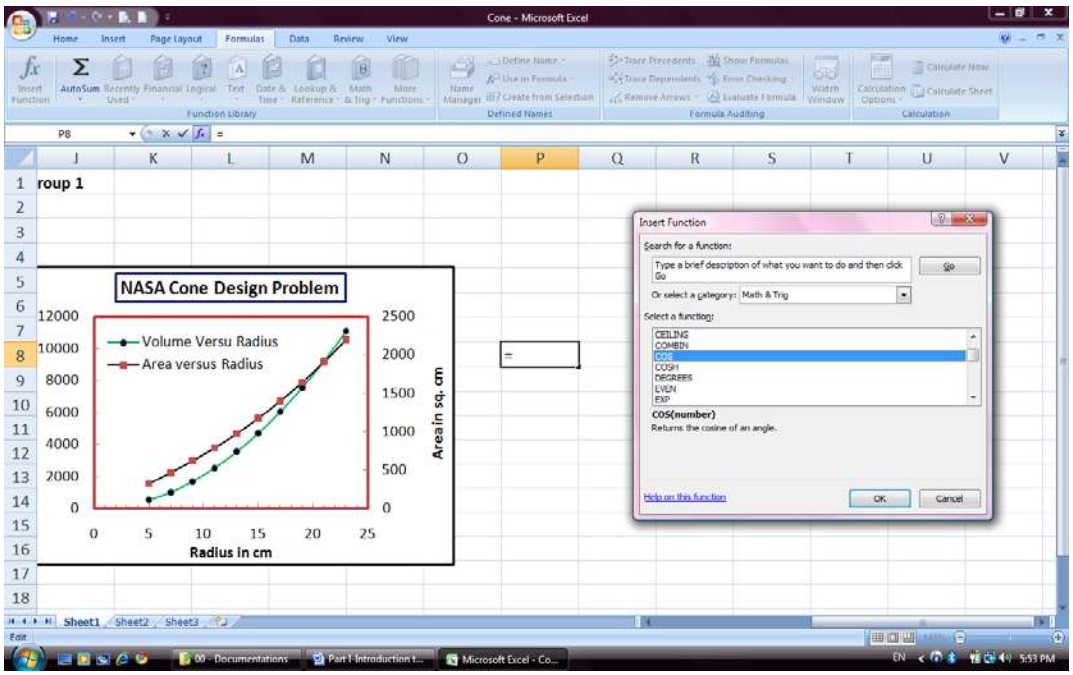
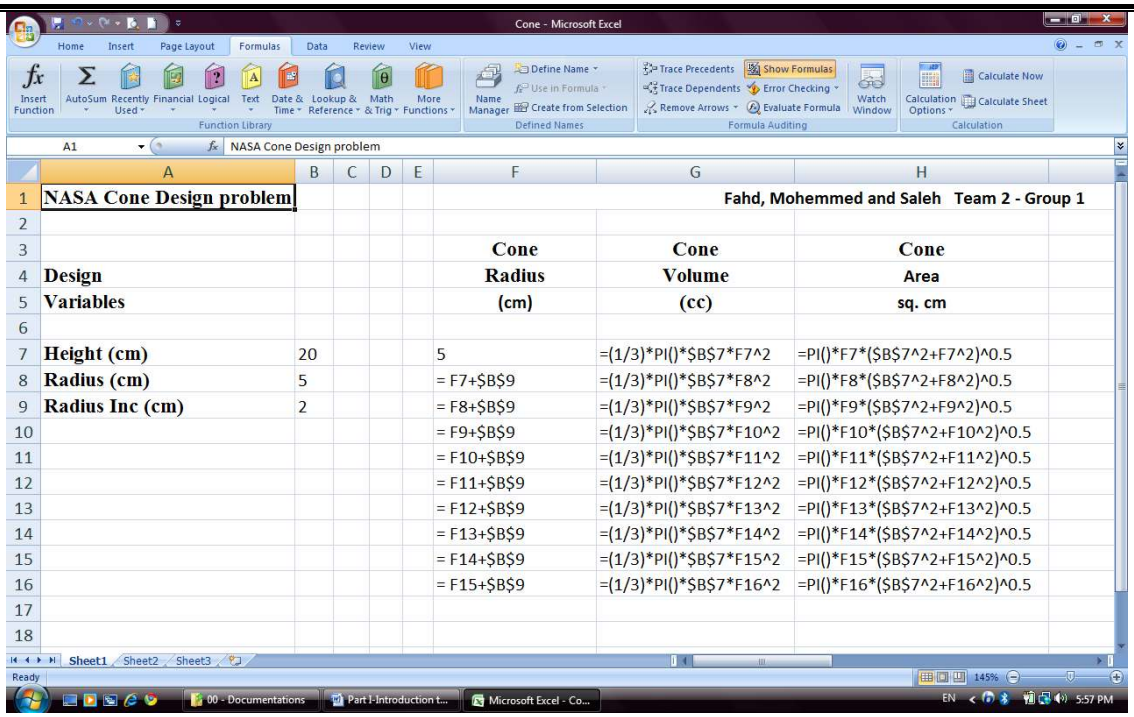


Figure 26B Choosing the cosine function

Trouble Shooting, Viewing & Printing Formulas

Testing, validating or calibrating a spreadsheet model is an essential task in developing correct, robust, reliable, and useful models. Locating and correcting errors in spreadsheets can be facilitated by viewing and/or printing the formulas in your spreadsheet. This process destroys much of the formatting you've created, so save your work before viewing the formulas. It is often convenient to save the viewed formula version under a different filename.

From the **Formula Tool Bar Menu**, using the left mouse button click on the button of **Show Formulas**. The formulas are now displayed in the cells rather than the values (see Figure 27)



	A	B	C	D	E	F	G	H
1	**NASA Cone Design problem**						**Fahd, Mohemmed and Saleh Team 2 - Group 1**	
2								
3						**Cone**	**Cone**	**Cone**
4	**Design**					**Radius**	**Volume**	**Area**
5	**Variables**					**(cm)**	**(cc)**	**sq. cm**
6								
7	**Height (cm)**	20				5	=(1/3)*PI()*$B$7*F7^2	=PI()*F7*($B$7^2+F7^2)^0.5
8	**Radius (cm)**	5				= F7+$B$9	=(1/3)*PI()*$B$7*F8^2	=PI()*F8*($B$7^2+F8^2)^0.5
9	**Radius Inc (cm)**	2				= F8+$B$9	=(1/3)*PI()*$B$7*F9^2	=PI()*F9*($B$7^2+F9^2)^0.5
10						= F9+$B$9	=(1/3)*PI()*$B$7*F10^2	=PI()*F10*($B$7^2+F10^2)^0.5
11						= F10+$B$9	=(1/3)*PI()*$B$7*F11^2	=PI()*F11*($B$7^2+F11^2)^0.5
12						= F11+$B$9	=(1/3)*PI()*$B$7*F12^2	=PI()*F12*($B$7^2+F12^2)^0.5
13						= F12+$B$9	=(1/3)*PI()*$B$7*F13^2	=PI()*F13*($B$7^2+F13^2)^0.5
14						= F13+$B$9	=(1/3)*PI()*$B$7*F14^2	=PI()*F14*($B$7^2+F14^2)^0.5
15						= F14+$B$9	=(1/3)*PI()*$B$7*F15^2	=PI()*F15*($B$7^2+F15^2)^0.5
16						= F15+$B$9	=(1/3)*PI()*$B$7*F16^2	=PI()*F16*($B$7^2+F16^2)^0.5
17								
18								

Figure 27 Excel formula sheet

If the entire formula is not displayed in the cells, use the cell formatting or using the mouse do the appropriate actions to display the entire formula (Figure 27). If you again click on button of **Show Formulas,** numerical values will be displayed again.

Printing

To print, from the Office Button select **Print** and then click the **Print Preview** button. Inside the print preview click **Page Setup**. You can change your page to be printed vertical or landscape depending on how your spreadsheet is set up. Once you have sized your document properly, click **OK**.

Now you can select **Print**, the default (and the setting you will most often use) is to print the **active sheet**. If you have more than one sheet in your workbook, you may wish to print the **entire workbook**. If you have only one part of your spreadsheet that you wish to print (just the formulas), you can highlight that region (before going to **Print** on the pull down menu) and then clicking on **Selection** when you go to **Print**.